INCH-POUND

MIL-G-18015B(SH)
2 May 1994
SUPERSEDING
MIL-G-18015A(SHIPS)
15 March 1957
(See 6.7)

MILITARY SPECIFICATION

GRATINGS, METAL, OTHER THAN BAR TYPE (SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aluminum and steel floor gratings other than bar types for shipboard use.

1.2 Classification. Gratings shall be of the following types and classes, as specified (see 6.2):

Type I - Steel, heavy
  Class 1 - Galvanized
  Class 2 - Ungalvanized
  Class 3 - Corrosion resistant
Type II - Aluminum, heavy
Type III - Steel, light
  Class 1 - Galvanized
  Class 2 - Ungalvanized
  Class 3 - Corrosion resistant
Type IV - Aluminum, light

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5670
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-A-200	- Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification For.
QQ-A-200/4	- Aluminum Alloy 5083, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-200/5	- Aluminum Alloy 5086, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-200/6	- Aluminum Alloy 5454, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-200/7	- Aluminum Alloy 5456, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-200/8	- Aluminum Alloy 6061, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-200/9	- Aluminum Alloy 6063, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-225	- Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes; Rolled, Drawn, or Cold Finished; General Specification For.
QQ-A-225/7	- Aluminum Alloy 5052, Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished.
QQ-A-225/8	- Aluminum Alloy 6061, Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished.
QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet: General Specification For.
QQ-A-250/6	- Aluminum Alloy 5083, Plate and Sheet.
QQ-A-250/7	- Aluminum Alloy 5086, Plate and Sheet.
QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet.
QQ-A-250/9	- Aluminum Alloy 5456, Plate and Sheet.
QQ-A-250/10	- Aluminum Alloy 5454, Plate and Sheet.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-S-766	- Steel Cres 304, Plate and Sheet.

MILITARY

MIL-M-17194	- Metal, Expanded, Steel.

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STANDARDS

MILITARY

MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 153 - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware. (DOD adopted)
A 167 - Standard Specification for Stainless and Heat-Resisting Chromium Nickel Steel Plate, Sheet, and Strip. (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Steel. Types I and III gratings shall be made of either carbon steel rolled, or drawn, or high tensile alloy steel, rolled or drawn, or cres 304 rolled, or drawn as specified in QQ-S-766.

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3.1.2 Aluminum. Types II and IV gratings shall be made of material meeting the chemical requirements of QQ-A-200 and QQ-A-200/4 through QQ-A-200/9, QQ-A-225, QQ-A-225/7 and QQ-A-225/8, QQ-A-250 and QQ-A-250/6 through QQ-A-250/11. When the material is not specified, the manufacturer may furnish any of these materials (see 3.2 and 6.2).

3.2 Construction. The grating panel sizes shall be as specified (see 6.2). Where nonrectangular shapes or cutouts in rectangular panels are required, they shall be as shown on the drawings as specified, (see 6.2). The edges of nonrectangular shapes and cutouts where more than 4 bearing bars or channels are included, or where the expanded metal is cut, shall be finished with banding bars of the same depth as the bearing bars. The banding bars when required shall be welded to the bearing bars or channels of the grating at least every fourth bearing bar or channel and in every corner. Intervening bearing bars or channels shall be tack welded. The ends of 6000 series aluminum grating panels shall be finished or banded by means of spacers or bridges to preclude welding. The 5000 series aluminum may be welded.

3.2.1 Clamps and fastenings. Suitable clamps or other fastenings shall be provided when specified, (see 6.2). Such clamps or fastenings shall be of the same material and be finished in the same manner as the gratings, shall not extend above the surface of the gratings, and shall be of such construction as to provide means for holding and securing the gratings to the supporting structure, with adequate simplicity of installation and for removal and replacement in service.

3.2.1.1 Thickness and spacing. The thickness and spacing of all bearing bars or channels and the depth, thickness, and spacing of all cross or rectilinear bars or expanded metal and frame shall be adequate to support the loads specified in table I without deflection due to bending, and without lateral displacement, warping, or distortion of any of the parts greater than 0.2 inch for steel, or 0.5 inch for aluminum, and without permanent set greater than 0.008 inch (see 4.3.4).

3.2.2 Strength and load. Gratings shall be of such construction (except cross-bar) as to meet the strength and load requirements of this specification (see 3.5). They may be of expanded metal, extruded and welded sections, or of other suitable construction (except cross-bar). Unless otherwise specified (see 6.2), gratings having serrated surface members will be permitted. Supports for gratings, if required, shall be of the dimensions specified (see 6.2).

3.2.2.1 Steel expanded metal. Steel expanded metal gratings shall meet the requirements of MIL-M-17194 and 3.5 of this specification, and such other requirements as may be specified (see 6.2).

3.2.3 Openings. Unless otherwise specified (see 6.2), the maximum dimensions of the openings shall be 15/16 by 3-7/16 inch as for expanded metal gratings and 1 by 4 inches for gratings of other types (see 3.3). Where specified, length of openings shall be based on load tests.

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3.3 Tolerances - general. Tolerances will be permitted as follows:

Length of bearing bars or channels	- plus or minus 1/16 inch in 6 feet.
Depth of bearing bars or channels	- plus or minus 1/32 inch (when measured from bottom of serrations when present).
Width of grating panels	- plus or minus 1/8 inch.
Openings	- plus or minus 10 percent.

A tolerance of plus 10 percent will be permitted on the maximum weight of any individual grating when banding bars or fasteners or both form a part of the grating.

3.4 Upper surface. The upper surface of gratings shall be free from sharp corners, burrs or projections which may tear or catch cloth or paper bags of provisions dragged over the gratings or interfere with normal cleaning methods.

3.5 Load. The load requirements of the gratings shall be as shown in table I.

TABLE I. Load requirements.

Steel				Aluminum		
Type	Design load per square foot	Test load1/	Maximum weight per square foot2/	Design load per square foot	Test load1/	Maximum weight per square foot2/
	Pounds	Pounds	Pounds	Pounds	Pounds	Pounds
I	300	900	5.00	---	---	----
II	---	---	----	300	900	5.00
III	175	525	4.40	---	---	----
IV	---	---	----	150	450	3.70

1/ Concentrated midway between support 2 feet long, on a 3-foot span (see 4.3.4).

2/ Finished, unbanded, including paint, bearing bars, or channels, and on types I and III, zinc-coating (galvanized).

3.6. Protective finish.

3.6.1 Types I and III, class 1.

3.6.1.1 Zinc-coating (galvanized). Unless otherwise specified, gratings shall be completely and evenly hot-dip zinc-coated (galvanized) after fabrication in accordance with ASTM A 153. The zinc-coating shall not separate from the basic metal when tested for adhesion as specified (see 4.3.3.1).

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3.7 Workmanship. Gratings shall be fabricated accurately and shall be free from warps, twists, and other defects affecting appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. For purposes of sampling, a lot shall consist of all gratings of the same type, class, and size offered for delivery at the same time.

4.2.1.1 Sample selection. Grating shall be selected in accordance with table II for the inspection of 4.3.1 through 4.3.3.

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TABLE II. Sampling for visual, weight, dimensional, and zinc adherence (regular and irregular shapes).

Lot size number of gratings or square feet	Sample size1/ number of gratings or square feet
2 to 5	All
6 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,000 to 35,000	35

1/ When ordered by square feet, no more than 5 square feet of the sample shall be in any one grating.

4.2.1.2 Sampling for load test. Sample gratings shall be selected from the lot in accordance with table III and subjected to the deflection and permanent set test of 4.3.4.

TABLE III. Sampling for test of deflection under load and permanent set.

Lot size Pieces or square feet	Regular shapes Sample pieces or square feet1/	Irregular shapes Sample pieces or square feet1/
3 to 40	3	3
41 to 65	3	4
66 to 110	3	5
111 to 180	4	7
181 to 300	5	10
301 to 500	7	15
501 to 800	10	20
801 to 1300	15	25
1301 to 3200	20	30
3201 to 8000	25	40
8001 to 22000	30	50
22001 to 110000	35	75
100001 to 550000	35	100
550001 to over	35	150

1/ When ordered by square feet, no more than 5 square feet of the sample shall be in any one unit.

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4.3 Examinations and tests.

4.3.1 Weight and dimensions. Each of the sample grating selected in accordance with 4.2.1.1 shall be examined to determine conformance with the weight and dimensional requirements of this specification (see 3.2.3, 3.3 and table I).

4.3.2 Protection finish. Each sample zinc-coated grating shall be surface examined after galvanizing to determine conformance with 3.6.1.1.

4.3.3 Zinc-coating.

4.3.3.1 Adhesion of coating. The sample gratings selected in accordance with 4.2.1.1 shall be struck a hard glancing blow with a straight peen or scaling hammer having all edges ground to a radius of not less than 3/64 inch and weighing approximately one pound.

4.3.4 Load. The sample gratings selected in accordance with 4.2.1.2 shall be supported by 2 supports with bearing width not to exceed 1 inch. These supports shall be placed 3 feet apart. The gratings shall be tested with the load specified for the type in accordance with table I, suspended from a bar placed midway between the supports. The bar shall not have a bearing width on the grating greater than 2 inches. Deflection and permanent set shall be determined by suitable gauges.

4.4 Inspection of packaging. Sample packs, and the inspection of the packaging (preservation, packing and marking) for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicablility of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.2.)

5.1 Packaging. Packaging (preservation and packing) shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Gratings, separated for type and class shall be packed for shipment in secured lifts on nominal 2-by-4 inch skids. Skidding and strapping shall be in accordance with MIL-STD-163. The maximum gross weight per lift shall not exceed 6,000 pounds for steel grating and 3,000 pounds for aluminum grating. Markings shall include bar coding.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The gratings covered by this specification are intended to be used for shipboard floor coverings.

6.2 Acquisition requirments. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of gratings required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Specification requirements for types II and IV gratings (see 3.1.2).
(e) Size of grating panels and drawings required (see 3.2.1.1).
(f) Whether clamps or fastenings are required (see 3.2.1.2). Welding is not allowed on 6000 series aluminum.
(g) Supports for gratings, if required (see 3.2.2).
(h) Dimensions of openings when larger than the maximum specified (see 3.2.3).
(i) Level of packaging required (see 5.1).

6.3 Superseding data. When bar type gratings are required, MIL-G-18014 should be used. When the intended use does not restrict the type of grating either MIL-G-18014 or this specification should be used.

6.4 Load test. If any lot is rejected for excessive deflection or permanent as specified in table IV, corrective action shall apply to all gratings in the lot.

TABLE IV. Test of deflection under load and permanent set.

Lot size	Regular shapes		Irregular shapes	
Pieces or square feet	Sample, pieces or square feet1/	Acceptance2/ constant (k)	Sample pieces or square feet1/	Acceptance2/ constant(k)
3 to 40	3	1.45	3	1.45
41 to 65	3	1.45	4	1.45
66 to 110	3	1.45	5	1.53
111 to 180	4	1.45	7	1.62
181 to 300	5	1.53	10	1.72
301 to 500	7	1.62	15	1.79
501 to 800	10	1.72	20	1.82
801 to 1300	15	1.79	25	1.85
1301 to 3200	20	1.82	30	1.86
3201 to 8000	25	1.85	40	1.89
8001 to 22000	30	1.86	50	1.93
22001 to 110000	35	1.89	75	1.98
110001 to 550000	35	1.89	100	2.00
550001 to over	35	1.89	150	2.03

1/ When ordered by square feet, no more than 5 square feet of the sample shall be in any one unit.

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2/ Acceptability criterion: The lot shall be accepted if $(U-\bar{X})$ s is greater than (k) the acceptance constant. U is the specification upper limits (see 3.2.1.3 and table I). $\bar{X}$ is the sum of the measurements like (ΣX) divided by the number of samples (n) or $\bar{X} = \Sigma X/n$. The lot standard deviations is equal to the following formula:

$$s = \sqrt{\frac{\Sigma X^2 - \frac{(\Sigma X)^2}{n}}{n - 1}}$$

Where: ΣX^2 is the sum of squared measurements.
$(\Sigma X)^2$ is the sum of measurements squared.

6.5 Sub-contracted material and parts. The preparation for delivery or packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment or material and lose their separate identity when the equipment or material is shipped.

6.6 Subject term (keyword) listing.

Aluminum
Corrosion resistant steel
Floor gratings
Galvanized steel

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-18015B(SH)	1994/05/02

3. DOCUMENT TITLE
GRATINGS, METAL, OTHER THAN BAR TYPE (SHIPBOARD USE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY TECHNICAL POINT OF CONTACT:TPOC:

a. NAME	b. TELEPHONE *(Include Area Code)*	
Bob Walls (Code 9716)	(1) Commercial	(2) AUTOVON
PLEASE ADDRESS ALL CORRESPONDENCE TO:	(215)897-7781	443-7781
Address *(Include Zip Code)* Commander, Naval Sea Systems Command SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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